Income Classification and Socioeconomic Segmentation

A Machine Learning and Business Analytics Study

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This report presents an end-to-end machine learning pipeline for predicting income levels using census data. The analysis combines statistical modeling, feature engineering, and unsupervised learning to generate both predictive and business insights relevant to risk segmentation and targeted decision-making.

Contents

1. Introduction: The Problem & Business Objective

Our client is interested in whether an individual earns above or below $50K annually using census-level demographic and socioeconomic attributes. Moreover, the client is interested in a segmentation model to group different subsets of the given population using said attributes.

From a business perspective, this problem is analogous to:

- Credit risk segmentation (approving vs rejecting applicants)
- Marketing targeting (high-income customer acquisition)
- Insurance pricing stratification

The business objective is two-fold:

1. **Income Classification:** Develop a supervised learning model that accurately predicts whether an individual earns less than $50,000 or more than $50,000 annually. This binary classification may serve as a proxy for identifying distinct economic tiers within the customer base.
2. **Customer Segmentation:** Build an unsupervised learning model that partitions individuals into meaningful behavioral and socioeconomic segments. These segments should capture latent structure in the population and may enable targeted marketing strategies.

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2. Exploratory Data Analysis (EDA)

2.1 Dataset Structure and Feature Semantics

This dataset is derived from census-based demographic and employment records compiled to support downstream decision-making for a retail client seeking to better understand and segment the U.S. population for targeted marketing strategies. The underlying data originates from large-scale survey collection efforts designed to capture socioeconomic characteristics across households.

The dataset contains mixed-type variables:

- **Continuous numeric variables** (income-related signals, work hours, capital gains/losses)
- **Categorical demographic variables** (education, occupation, marital status)
- **Survey weights** indicating the relative distribution of people in the general population that each records represents due to stratified sampling.

Business Insight: Survey weights imply that this dataset is not purely observational but statistically representative of a population. This is important because it helps level out the bias towards overrepresented demographics. In a production setting, this over-representation could directly translate into misallocation of marketing spend and systematically skewed targeting strategies.

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2.2 Target Variable Distribution

The target variable is binary:

- 0: Income $\leq$ \$50K
- 1: Income $>$ \$50K

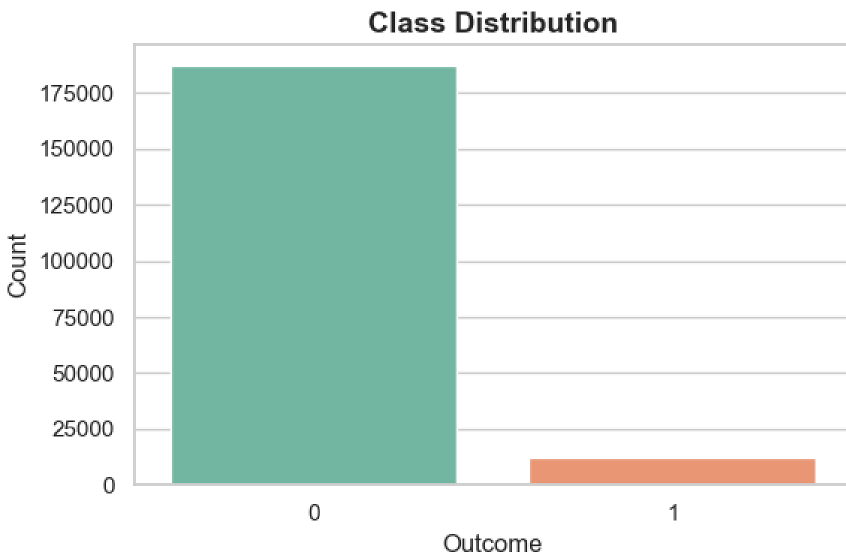

Figure 1: The class distribution for the target variable. The distribution is highly imbalanced, with a dominant low-income class.

Due to the data imbalance, this could lead to some technical implications. First, accuracy becomes a misleading metric due to the fact that the model could naively predict all minority class and achieve a high accuracy. Thus, an ROC-AUC score will be used as it captures ranking quality and is more appropriate within a business context. Next, the imbalance in outcome could lead to an imbalance in the train/test split within a modeling context. As a result, stratified sampling will be required in order to preserve class ratios within the training and testing data.

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2.3 Feature Distribution and Outlier Structure

Table 1: Summary Statistics for Income-Related Financial Variables

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Wage per Hour	55.43	274.90	0.00	0.00	0.00	0.00	9999.00
Capital Losses	37.31	271.90	0.00	0.00	0.00	0.00	4608.00
Capital Gains	434.72	4697.53	0.00	0.00	0.00	0.00	99999.00
Dividend Income	197.53	1984.16	0.00	0.00	0.00	0.00	99999.00

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The summary statistics within Table 1 reveal strong skewness in:

- Capital gains
- Capital losses
- Dividend income
- Wage per hour

These variables seem to be heavily right-skewed, with most observations clustered at zero and a small number of extreme values driving the upper tail. This pattern indicates strong zero-inflation and the presence of outliers that can distort downstream analysis if left unaddressed. From a business standpoint, these features capture rare but financially significant behaviors such as investment activity and realized capital gains, which tend to be strong separators between lower and higher wealth segments.

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2.4 Class-Conditional Feature Separation

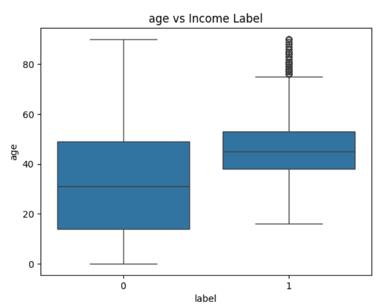

Figure 2: Boxplot of Age vs. Label

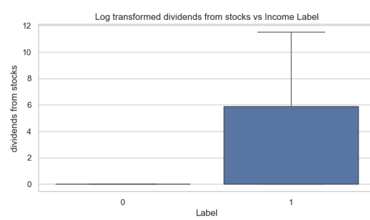

Figure 3: Boxplot of Dividends from Stocks vs. Label

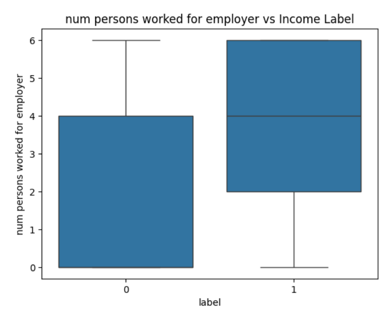

Figure 4: Boxplot of Workplace Size vs. Label

Boxplots (as shown in figures 2, 3, and 4) grouped by label show that:

- High-income individuals have higher dividends from stocks and have a larger workplace.
- Low-income individuals tend to be younger.

This indicates strong nonlinear separability in feature space.

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2.5 Categorical Feature Informativeness

Many of the categorical features provided strong signals for income, indicating that income separation is not driven solely by continuous financial variables but is also deeply embedded in demographic and occupational structure.

Target-mean encoding (probability of income >50K per category) reveals (as shown in figures 5, 6, and 7):

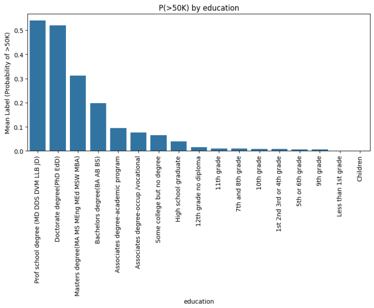

Figure 5: Mean-encoded Label by Education Level

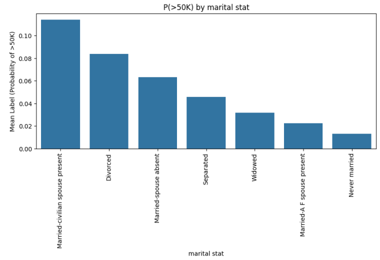

Figure 6: Mean-encoded Label by Marital Status

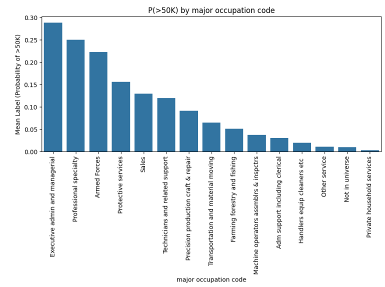

Figure 7: Mean-encoded Label by Occupation

- Strong signal in education level
- Marital status effects
- Occupation type

From a modeling perspective, this suggests that categorical variables such as education level, occupation type, marital status, and industry classification carry high mutual information with the target label. In other words, the probability of earning above 50K varies meaningfully across discrete groups rather than being uniformly distributed across the population. From a business standpoint, this is particularly important for marketing segmentation.

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3. Data Cleaning and Feature Engineering

3.1 Missing Value Strategy

Missing values encoded as “?” were converted to NaN and treated using:

- Dropping columns with >40% missingness
- Imputing remaining missing values as “Unknown”

The reason why I decided to drop columns with greater than 40% missingness is because a high amount of missing data introduces bias and instability. Moreover, for columns with less than 40% missingness, I decided to impute them as ”Unknown” to preserve signal rather than dropping those rows or imputing some other artificial structure.

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3.2 Zero-Inflation Handling

Key financial variables are heavily zero-inflated.

We apply:

- Binary indicator flags (presence vs absence)
- Log transformation for non-zero values

The reason for creating a flag is that this separates structure zeroes from magnitude effects. Meanwhile, the log-transformation for non-zero values reduces skewness and likely improves linear separability. Because logistic regression is our model of choice, the log-transformation will be helpful in reducing the effects of the outliers (which are acknowledged but will not be removed).

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4. Predictive Modeling

4.1 Model Choice: Logistic Regression

Logistic Regression is a supervised classification algorithm used to model the probability that a given observation belongs to a particular class. In this case, it estimates the probability that an individual earns more than $50K by applying a linear combination of input features, which is then passed through a sigmoid function to map outputs into the range [0,1].

Why this model?

- High interpretability through coefficient analysis (critical for business deployment)
- Stable coefficients for feature importance
- Strong baseline for structured tabular data

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4.2 Preprocessing Pipeline

We apply:

- One-hot encoding for categorical variables
- Standardization for numeric variables

Standardizing the numerical features serves both a technical and business purpose. From a technical standpoint, it prevents variables with large numeric ranges from dominating the optimization process simply because of their scale, ensuring that the regularization penalty is applied evenly across all features. This is particularly important in logistic regression, where coefficient magnitude is directly influenced by feature scale and can otherwise lead to distorted importance estimates.

From a business perspective, standardization makes the resulting model coefficients directly comparable across variables. This will allow the client to interpret which factors most strongly influence income predictions in a consistent and meaningful way, supporting clearer decision-making around which markets to target.

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4.3 Regularization and Hyperparameter Tuning

Regularization helps prevent the model from naively memorizing or adhering to the training data (aka. overfitting). Moreover, hyperparameter tuning is a framework that allows us to get the best estimate for which kind of regularization to use as some work better than others.

We test:

- L2 (ridge regression)

- L1 (lasso regression for feature selection)
- Elastic Net (hybrid sparsity + stability, combines L1 + L2)

Through a grid search, these 3 regularization techniques will be tested, as well as different C values (which allows one to control "how much" of the regularization to use).

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5. Model Evaluation

5.1 Training vs Validation Behavior

Regularization	C	Train ROC-AUC	Validation ROC-AUC
L2	0.01	0.948246	0.945438
L2	0.10	0.948248	0.945363
L2	1.00	0.948228	0.945386
L2	10.00	0.948277	0.945409
L1	0.01	0.894280	0.891515
L1	0.10	0.898392	0.893846
L1	1.00	0.897062	0.894470
L1	10.00	0.874789	0.866207
Elastic Net	0.01	0.894921	0.891579
Elastic Net	0.01	0.899398	0.898178
Elastic Net	0.10	0.898472	0.898141
Elastic Net	0.10	0.894238	0.893659
Elastic Net	1.00	0.900384	0.894884
Elastic Net	1.00	0.906507	0.901931

Table 2: Grid Search Results Across Different Logistic Regression Regularization Strategies

We observe:

- Increasing regularization reduces overfitting (though, marginally)
- Optimal C balances bias-variance tradeoff

Interpretation: We see that L2 regularization dominates and is the most appropriate regularization for this model. L2 is generally used to produce stable coefficients, and is helpful when there are potentially correlated variables in the dataset. Moreover, L2 is strongest when the features used within the model are at the very least moderately informative.

5.2 Final Model Results

Since the ROC-AUC metric is used to evaluate the model, it is appropriate to produce a corresponding ROC-AUC curve and its score for the best performing model.

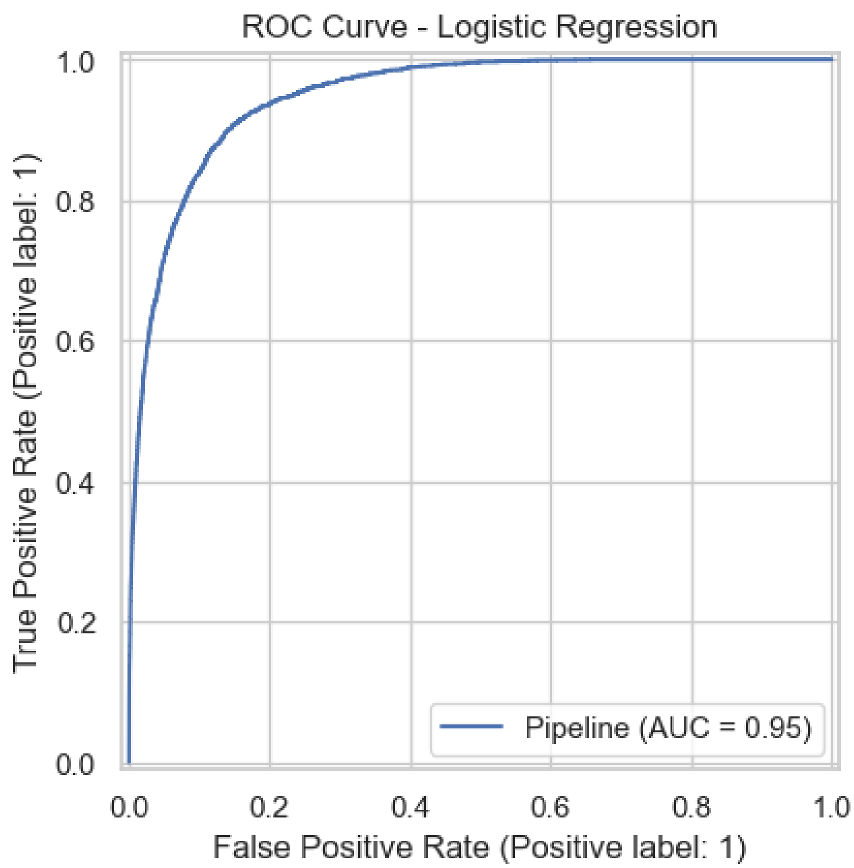

Figure 8: The ROC-AUC curve and score for the corresponding best performing model.

Metric	**Value**
Accuracy	0.9539
Precision	0.7318
Recall	0.4055
F1 Score	0.5218
ROC-AUC	0.9488

Table 3: Logistic Regression Test Set Performance Metrics

The Logistic Regression model achieved a strong overall accuracy of 95.39%, though accuracy is less informative due to class imbalance in the dataset. The model obtained a precision of 73.18%, indicating that predicted high-income individuals are typically classified correctly, which is valuable for reducing inefficient marketing outreach. The recall score of 40.55% suggests the model is relatively conservative and does not capture all high-income individuals, instead prioritizing prediction reliability. This tradeoff is reflected in the F1 score of 0.5218. Finally, the ROC-AUC score of 0.9488 demonstrates excellent class separation capability, indicating that the model is highly effective at ranking higher-income individuals above lower-income individuals for targeting and segmentation purposes. Depending on the needs of the client, the recall or precision score can be tuned to meet the demands.

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6. Unsupervised Learning: Customer Segmentation

6.1 KMeans Clustering

K-Means clustering is an unsupervised machine learning algorithm used to identify naturally occurring groups within data. The algorithm partitions observations into K clusters by minimizing the distance between data points and their assigned cluster centroids. It is an effective way to identify latent socioeconomic groups.

Why clustering?

- Reveals hidden structure beyond labels
- Enables business segmentation strategies

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6.2 Elbow Method

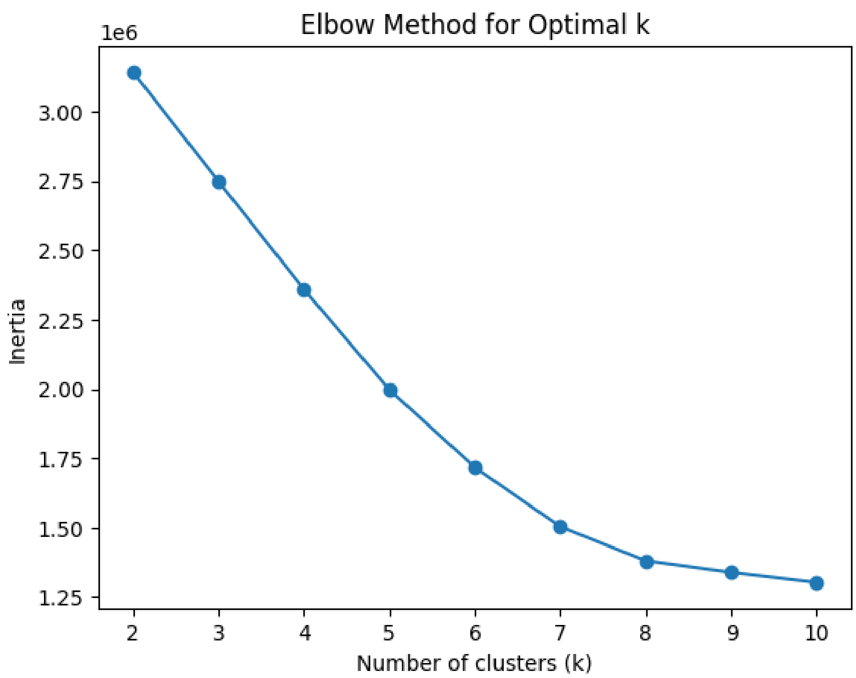

Figure 9: Elbow Plot for the Number of Clusters to Choose.

We select $k = 4$ based on inertia minimization through a loose use case of the elbow method. Practically, 8 clusters is usually too granular for segmentation analysis, but the elbow method tells us to use 8. Since the elbow method is largely heuristic and less clusters are better within a business context (more likely to be actionable), I opt to keep 4 clusters especially since its inertia value of 2.25 is not too far off from the 1.5 inertia value of having 8 clusters.

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6.3 Cluster Profiling

Based on the numerical and categorical feature distributions, each cluster shows a distinct socioeconomic profile, with clear differences in income likelihood:

- **Cluster 0 (32.7% high income)**: This group consists of older, financially established individuals with high investment activity (capital gains and dividends), higher education levels, and strong representation in professional and managerial roles. It is predominantly male and married, suggesting stable, high-earning households.
- **Cluster 1 (0.4% high income)**: This cluster represents a younger or economically inactive population with minimal work participation, almost no capital or wage activity, and a large "not in universe" occupation share. It is mostly never-married and lower income overall.
- **Cluster 2 (9.4% high income)**: This is a working middle-class segment with high employment intensity (most weeks worked), broad occupational diversity, and income driven mainly by wages rather than investments. It is fairly balanced in gender and represents stable earners.
- **Cluster 3 (29.9% high income)**: This group is similar to Cluster 0 in age, education, and occupation mix, but shows more capital loss activity and slightly different financial behavior patterns. It still represents a relatively high-income professional segment with strong labor participation.

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7. Conclusion

This project was designed to support the client's marketing objective of identifying high-income individuals and segmenting the broader population into actionable groups. All modeling choices were made with a focus on interpretability, deployability, and direct business usability.

For the supervised task, Logistic Regression with L2 regularization was selected as the final model due to its strong predictive performance (ROC-AUC = 0.9488) and, more importantly, its interpretability. A key advantage of this approach is that model coefficients can be directly inspected to understand which variables most strongly drive income predictions. Positive coefficients indicate features associated with higher income probability, while negative coefficients indicate the opposite, allowing the client to clearly identify key socioeconomic drivers of high-income status.

The model is best used as a ranking tool, where predicted probabilities guide marketing prioritization rather than strict classification. For segmentation, K-Means clustering with four clusters was chosen as a deliberate tradeoff between statistical granularity and business interpretability, ensuring each group remains actionable for marketing strategy.

Overall, the recommended approach is a two-layer system: use Logistic Regression for targeted outreach prioritization and clustering for population segmentation. Together, they provide both predictive power and clear business interpretability.